

SAR-to-Optical image Translation

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Abstract

Cloud cover remains a persistent challenge in optical remote sensing, limiting the usability of optical satellite imagery for continuous Earth observation. Synthetic Aperture Radar (SAR) data, in contrast, provides cloud-penetrating, all-weather imaging but lacks the spectral and visual richness of optical observations. Bridging these complementary modalities, this study investigates SAR-to-optical image translation using the Pix2Pix conditional generative adversarial network (cGAN). While existing research predominantly focuses on reconstructing only the visible (RGB) or near-infrared bands, this work employs the winter subset of the SEN12-MS dataset to address the full spectral range. The objectives are threefold: (i) to validate SAR-to-optical translation across all 13 Sentinel-2 spectral bands; (ii) to assess the reliability and reconstructability of each individual band; and (iii) to evaluate the performance of the translation model for cloud removal. Experimental results show that the model effectively learns the SAR-to-optical mapping and achieves high reconstruction quality across all spectral bands, though bandwise analysis reveals that reconstruction accuracy varies with spectral characteristics. When applied to the SEN12-MS-CR dataset, the model successfully generates cloud-free optical imagery that closely matches reference data, achieving performance competitive with state-of-the-art models such as DiffCR. Overall, the findings confirm the viability of SAR-to-optical translation for producing spectrally consistent, cloud-free optical imagery, thus enhancing the temporal continuity of Earth observation data. Two ablation studies further analyze the impact of different loss functions and the exclusion of 60 m bands. Additionally, based on data analysis, a new per-band clipping strategy for optical data is proposed.

Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article; yet reasonably common within the subject discipline.)

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2. Introduction

Earth observation has become an indispensable tool for understanding and monitoring the planet's dynamic processes. Optical and radar remote sensing represent two complementary modalities at the core of modern monitoring systems. Optical sensors, such as the Multi-Spectral Instrument (MSI) onboard Sentinel-2, provide rich spectral and visual information essential for applications ranging from agricultural monitoring and land-cover classification to disaster response and forest monitoring [1]. However, these sensors are inherently constrained by atmospheric conditions, particularly cloud cover. Global estimates suggest that average cloud cover exceeds 66% [4–6], with about 55% over land surfaces [6], causing considerable data gaps in both spatial and temporal domains. For applications requiring consistent time series, e.g., agricultural monitoring, or where a specific scene must be observed at a given time, e.g., disaster monitoring, cloud cover represents a serious limitation [6]. The diversity of clouds—including thin and thick clouds as well as haze—together with the wide range of occlusion scenarios and their uneven distribution, poses an additional challenge for image reconstruction and the generalizability of cloud removal techniques [7]. In contrast, Synthetic Aperture Radar (SAR) sensors, operating in the microwave domain, such as the C-band instrument on Sentinel-1 mission offer all-weather, day-and-night imaging capabilities independent of solar illumination or cloud interference. However, the backscatter-based nature of SAR imagery introduces challenges related to speckle noise, geometric distortions, and a lack of intuitive spectral color information [2,3], necessitating advanced interpretation skills.

To bridge the gap between these two sensing modalities, SAR-to-optical image translation has emerged as a powerful generative approach. It aims to synthesize optical-like, cloud-free imagery from SAR data, combining the interpretability of optical observations with the reliability of radar acquisitions. Recent advances in generative artificial intelligence (GenAI), particularly in conditional generative adversarial networks (cGANs) [8] and diffusion models [9], have made it possible to learn complex mappings between SAR and optical domains with visually plausible results [10]. These developments have opened new pathways for applications in land-cover classification, vegetation monitoring, disaster response, and particularly, cloud removal.

Despite the rapid progress in this field, a significant gap remains in the literature. Most existing studies have focused primarily on reconstructing the visible RGB subset of optical imagery such as [2,10–13], with a few extending to the Near-Infrared (NIR) range such as [14,15]. Consequently, the full multispectral potential of missions such as Sentinel-2 remains largely underexplored in the context of generative reconstruction. Critical bands for vegetation analysis (Red-Edge) and soil moisture monitoring (SWIR) are rarely addressed.

Furthermore, while SAR-to-optical translation is often evaluated qualitatively, systematic assessments of its reliability across individual spectral bands remain scarce. Finally, although the approach shows promise for cloud removal, its effectiveness in this context has not yet been comprehensively validated across the full spectral range.

Accordingly, this study investigates the use of generative models for translating Sentinel-1 SAR imagery into the full multispectral stack of Sentinel-2 imagery. The objectives of this work are threefold:

1. To validate SAR-to-optical image translation across all 13 Sentinel-2 spectral bands, moving beyond standard RGB constraints.
2. To systematically assess the reliability and reconstructability of each individual spectral band.
3. To evaluate the performance of the translation model for the practical application of cloud removal.

By addressing these objectives, this work aims to provide a comprehensive and quantitative
understanding of SAR-to-optical translation as a multimodal learning problem and to clarify
its potential and limitations for enhancing the temporal continuity of Earth observation
data.

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3. Related Work

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3.1. Generative Adversarial Network

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This section may be divided by subheadings. It should provide a concise and precise description of the experimental results, their interpretation as well as the experimental conclusions that can be drawn.

5.1. Subsection

5.1.1. Subsubsection

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- Second bullet;
- Third bullet.

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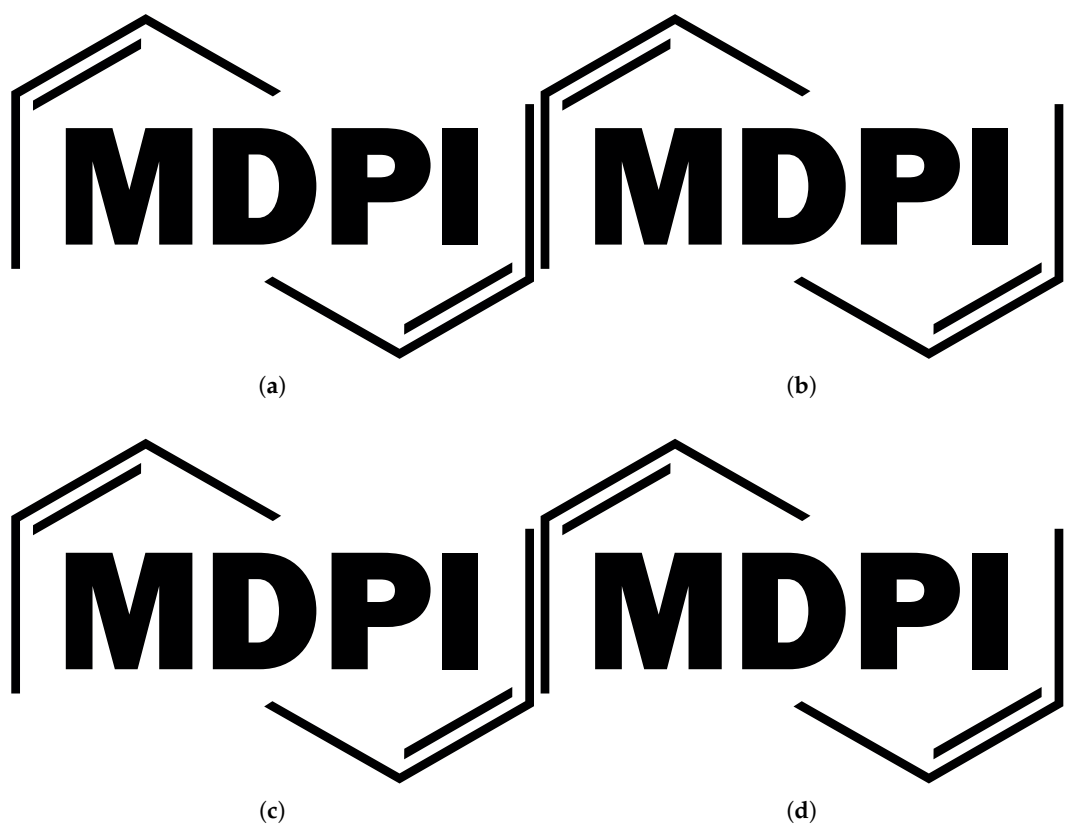


Figure 2. This is a wide figure. Schemes follow the same formatting. If there are multiple panels, they should be listed as: (a) Description of what is contained in the first panel. (b) Description of what is contained in the second panel. (c) Description of what is contained in the third panel. (d) Description of what is contained in the fourth panel. Figures should be placed in the main text near to the first time they are cited. A caption on a single line should be centered.

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Entry 2	Data	Data	Data
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	Data	Data	Data

* Tables may have a footer.

Text.

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This is the example 1 of equation:

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$$a = 1,$$

(1)

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the text following an equation need not be a new paragraph. Please punctuate equations as regular text.

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This is the example 2 of equation:

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$$a = b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z$$

(2)

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Please punctuate equations as regular text. Theorem-type environments (including propositions, lemmas, corollaries etc.) can be formatted as follows:

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Theorem 1. *Example text of a theorem.*

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The text continues here. Proofs must be formatted as follows:

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Proof of Theorem 1. Text of the proof. Note that the phrase “of Theorem 1” is optional if it is clear which theorem is being referred to. □

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The text continues here.

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Authors should discuss the results and how they can be interpreted from the perspective of previous studies and of the working hypotheses. The findings and their implications should be discussed in the broadest context possible. Future research directions may also be highlighted.

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This section is not mandatory, but can be added to the manuscript if the discussion is unusually long or complex.

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Abbreviations

The following abbreviations are used in this manuscript:

- MDPI Multidisciplinary Digital Publishing Institute
- DOAJ Directory of open access journals
- TLA Three letter acronym
- LD Linear dichroism

Appendix A

Appendix A.1

The appendix is an optional section that can contain details and data supplemental to the main text—for example, explanations of experimental details that would disrupt the flow of the main text but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data are shown in the main text can be added here if brief, or as Supplementary Data. Mathematical proofs of results not central to the paper can be added as an appendix.

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Entry 2	Data	Data

Appendix B

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